Computational Thinking and Artificial Intelligence

Class 8

Student Handbook

CENTRAL BOARD OF SECONDARY EDUCATION

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PREFACE

The National Education Policy (NEP) aims to position India as a leader in emerging knowledge fields by integrating technologies like AI, Machine Learning, Big Data and Computational Thinking into school education. It promotes technology-enabled, interactive and gamified learning using tools such as Augmented Reality (AR), Virtual Reality (VR), and virtual labs to foster creativity, problem-solving, and interdisciplinary exploration. NCFSE 23 carries this recommendation further for implementation.

While Artificial Intelligence (AI) is an important requirement, Computational Thinking (CT) should be a broader skill, developing a foundation for learning AI. It can cover various aspects like Cybersecurity, basic networking, etc. Hence, CBSE approaches this by integrating Computational Thinking with AI and other technological advancements, without dependence on any platform.

Learners engage with problems involving powers and number systems, proportional reasoning, geometric configurations and structured distributions, requiring decomposition of multi-variable scenarios, identification of complex patterns and design of stepwise algorithms under constraints. The Artificial Intelligence component deepens understanding of the AI project lifecycle, data-driven decision-making and ethical considerations such as bias and fairness, enabling students to critically analyse how data and models influence outcomes. The document also provides pedagogical guidance, resources, and assessment support aligned with NEP 2020 for effective classroom implementation.

TEAM CBSE

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Introduction

Computational Thinking (CT) is a problem-solving approach that comprises Decomposition, Pattern Recognition, Abstraction, Algorithm Design, Data Analysis and Troubleshooting. Computational Thinking Skills involve solving complex problems that promote thinking skills such as critical & creative thinking, abstraction and pattern recognition, as well as algorithmic thinking. Problem identification and problem solving necessitate the application of multidisciplinary understanding for creating effective solutions.

Artificial intelligence (AI) is a cutting-edge technology that empowers machines and computers to perform tasks that usually require mimicking human intelligence. These machines can perform complex thinking processes such as data analysis, pattern recognition, prediction of trends, solving problems and decision making. Thus, AI involves simulating cognitive processes associated with human intelligence and is widely applicable in various sectors such as banking, healthcare, defense, education, entertainment, agriculture and others, for processing information, solving intricate problems and for planning.

The National Education Policy (NEP) aims for India to emerge as a global leader in new emerging knowledge domains such as artificial intelligence, machine learning, data analytics, 3-D machining etc. To realise this goal, the policy suggests teaching students' mathematics and computational thinking, along with new subjects like artificial intelligence, machine learning, and data science during their school education. The policy also focuses on technology-enabled learning and classrooms by using tools like artificial intelligence, machine learning and adaptive testing to create knowledge.

The National Curriculum for School Education draws from this policy aspiration and emphasises the need to introduce these emerging domains of study and technologies in the school curriculum. It recommends inclusion of subjects such as design thinking, augmented reality, virtual reality, artificial intelligence, and computational thinking. Additionally, it promotes the use of gamified content, interactive content, and immersive experiences (such as AR, VR or virtual labs) to enhance student learning. In a variety of subjects, including design, music, art and sciences, these resources support students in knowledge creation and exploration, and development of capacities such as problem-solving, critical and creative thinking.

CBSE, under the aegis of the Department of School Education and Literacy, Ministry of Education, Govt. of India, is implementing a Curriculum on Computational Thinking and Artificial Intelligence (CT & AI) to inculcate AI-readiness in school students. This curriculum will be implemented from classes 3rd to 8th, in the session 2026-27, and aims to develop AI-Ready learners, by focusing on Computational Thinking Skills. The AI-readiness, so inculcated through CT Skills, will help develop the capacities of learners to use computational thinking, such as logical thinking, problem solving, pattern recognition, and so on, and understand the role and use of Artificial Intelligence in daily life. The Curriculum aims to build strong foundations in computational thinking, digital literacy and responsible use of technology, along with nurturing innovation, critical thinking, and ethical decision-making capacities.

1. **Relevance: Importance of introducing Computational Thinking (CT) and Artificial Intelligence (AI)**

- **Preparing for the future:** To contribute to the world of work in modern societies, individuals need capabilities such as problem solving, using data effectively, identifying patterns and applying AI ethically for various purposes in life.
- **Holistic Development:** Study of CT and AI contributes to development of reasoning, logical thinking, creative problem-solving skills, critical thinking, and ethical decision-making abilities, leading to individual flourishing and the creation of responsible digital citizens.
- **Interdisciplinary Relevance:** Embedding CT and AI concepts helps students develop an integrated view of the world by connecting various disciplines such as Mathematics, Science, and Humanities, showing that knowledge is not compartmentalized.
- **Innovation and Entrepreneurship:** At its core, CT and AI are about solving problems and devising innovative solutions, which leads to an entrepreneurial and innovative mindset.
- **Ethical Awareness:** Study of CT & AI will sensitize learners about the misuse and bias, fairness, and inclusivity in AI systems.

2. **Objectives: (Curricular Goals)**

- **CG-1:** Develops skills and capacities of computational thinking, namely- decomposition, pattern recognition, data representation, generalisation, abstraction, and algorithms to solve problems where such techniques of computational thinking are effective.
- **CG-2:** Develop spatial and visual reasoning.
- **CG-3:** Gain foundational knowledge of AI, its types, and domains.
- **CG-4:** Understand key ethical terms such as bias and fairness in relation to AI.
- **CG-5:** Demonstrates proficiency to use Computer & other devices, computer applications for learning and practical purposes such as data analysis, preparation of visual representations and communication of ideas.

3. **Learning Outcomes:**

Computational Thinking (CT) Learning Outcomes

ABSTRACT THINKING

Students will be able to solve advanced, multi-layered problems involving abstract relationships and hidden structures, using:

- properties and relationships of numbers (powers, factors, remainders, divisibility)
- generalization across different number systems (decimal, binary, ternary, Roman, Chinese numerals)
- spatial visualization of 2D and 3D figures, including overlaps, intersections, and transformations
- logical interpretation of symbols, codes and operations representing numerical or algebraic ideas
- identification of essential information by ignoring irrelevant or misleading data

PATTERN RECOGNITION

Students will be able to identify, compare and extend complex patterns involving multiple simultaneous changes, formed using:

- Powers, exponents and numerical structures
- Relationships across different representations of the same number
- Geometric configurations and shape-based sequences
- Conditional patterns based on rules, constraints or dependencies
- Mixed patterns involving numbers, symbols, shapes, and movement

DECOMPOSITION

Students will be able to break down high-order logical problems into manageable components by:

- Separating given conditions, constraints, and goals
- Analyzing multi-step processes such as distribution, transfers and exchanges
- Breaking numerical expressions into simpler equivalent forms
- Interpreting tables, grids, networks and diagrams with multiple dependencies
- Structuring problems involving multiple variables, positions or cases

ALGORITHMIC THINKING

Students will be able to design, follow and evaluate multi-step logical procedures to solve problems involving:

- Rule-based transformations of numbers or symbols
- Stepwise movement on grids, tracks or paths with constraints
- Conditional instructions (if–then, either–or, must/must not)
- Sequential decision-making under given limitations
- Optimisation problems involving maximum or minimum outcomes

Artificial Intelligence (AI) Learning Outcomes

By the end of Grade 8, learners will be able to:

- Describe the stages of the AI project cycle as a stepwise structure (Define Problem, Collect Data, Test AI Tools, Reflect and Improve)
- Apply no-code tools to tackle real-world problems and reflect on their utility/effectiveness
- Explain how AI uses data, find and research sources of bias in datasets, and apply basic strategies to ensure fairness and inclusivity
- Recognize how bias in AI leads to unfair conclusions and realize the importance of accountability, privacy, and serving human interests
- Explain the uses of AI in daily life and understand AI as a specific type of algorithm that uses datasets, learning and prediction
- Analyse contributions of AI to fields like healthcare, automation, and education, understanding both benefits and risks
- Describe AI ethics as the values and guidelines that ensure AI is created and used responsibly

4. Mapped with NEP and NCF 2023:

- The National Education Policy (NEP) 2020 aims to position India as a leader in emerging fields by integrating AI, Machine Learning and CT into school education

- The National Curriculum Framework for School Education (NCF-SE), 2023 serves as the foundation for implementation, drawing Curricular Goals from the Aims of Education.
- Learning standards are designed as foundational capacities that are progressive, age-appropriate, and aligned to NCF-SE 2023.

5. Time Allocation:

The Middle Stage suggests 100 hours annually, allocated as follows for Grade 8:

- **Advanced CT Skills:** 40 hours per academic year
- **Introductory Concepts of AI:** 20 hours per academic year
- **Interdisciplinary Projects:** 40 hours total (20 hours for each of the two required projects)

6. Approach / Pedagogy:

- **Activity-Based:** Use of complex puzzles, riddles, and games to build on previous CT abilities
- **Experiential Learning:** Delivering fundamental AI concepts through explanations, demonstrations, and hands-on experience
- **Collaborative Work:** Organising group discussions, debates and collaborative projects that integrate CT & AI
- **Inquiry-Based:** Independent student activities such as data collection, organisation, analysis, and creation of diagrams/flow charts using digital tools or manually
- **Ethical Reflection:** Case studies and debates on the social impact and ethical use of AI.

7. Assessment:

Assessment is continuous, formative and competency-based, focusing on the ability to apply knowledge rather than rote memorization. Methods include:

- Written Tests and Practical Examinations
- Interactive Group Activities
- Thematic Projects and Reflective Journals
- Teacher Observation Journals and Group Discussions

8. CT and AI Transition:

Computational Thinking forms the intellectual backbone and foundation for learning AI. The curriculum follows a phased approach where CT skills—like breaking problems into parts and spotting patterns—build the cognitive structures necessary for students to eventually understand and create AI-driven solutions.

How to Use This Book?

PART-1 Computational Thinking

Part 1 of this handbook is designed as a companion to the Mathematics textbook and is intended to be used alongside regular classroom teaching. Since it follows the same chapter sequence, the Mathematics teacher can seamlessly integrate it into daily instruction. As concepts are introduced in class, the corresponding questions from this book can be used to deepen understanding and encourage application.

Before beginning a chapter, the teacher is encouraged to read and identify the underlying concepts required for each question and plan how to align them with classroom teaching. As these concepts are taught, the teacher can introduce the related 'thinking questions' to students. It is important to note that the questions in this book are thinking-based and designed to promote analysis, reasoning, and problem-solving.

Teachers should adopt a facilitative approach, guiding students through prompts and discussions rather than directly providing solutions. Students should be given time to think and attempt independently, followed by classroom discussions where different approaches are shared and explored.

Some chapters also include activities that build intuition and engagement. These should be conducted before attempting the questions, as they help students approach the problems with better understanding.

PART-2 Artificial Intelligence

Part 2 of the handbook provides a structured introduction to Artificial Intelligence (AI) as a technology that enables machines to learn from data, recognise patterns, and make decisions. The concepts of AI are presented using simple explanations and real-life examples from areas such as healthcare, education, transport, and communication.

Each chapter includes:

- Foundational understanding of AI concepts
- Real-life examples and applications of AI
- Introduction to key AI domains such as Data Science, Computer Vision, and Natural Language Processing
- Activities and data-based tasks
- Reflection on ethical use of AI

The AI content progresses from introduction to application, including introductory predictive techniques such as regression, classification, and clustering. The book emphasises ethical and responsible use of AI, including introduction to bias, fairness, privacy, and safe use of technology, enabling informed and thoughtful engagement with AI systems.

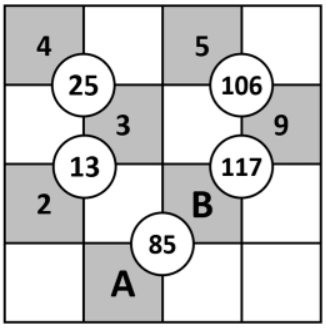
Teachers should approach the book with the mindset that the process of thinking is more important than arriving at the correct answer. Creating a safe and encouraging environment where students feel comfortable making mistakes, exploring multiple strategies, and expressing their reasoning is essential. The goal is to nurture confident, independent thinkers rather than focus solely on correctness.

PART-1

COMPUTATIONAL THINKING

Chapter 1: A Square and a Cube

1. **In the following grid, all the circles follow the same theme. What will be the value of A + B?**



a) 12 b) 14 c) 13 d) 15

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2. **Sam writes a list of natural numbers. The list has three perfect cubes and three perfect squares. If no number in the list has more than two digits, what is the MINIMUM number of distinct numbers he must have written?**

a) 3 b) 4 c) 5 d) 6

---

3. **If AB is a two-digit number whose cube is in the form of a 4-digit number "__ __ __ C" such that A < C < B, how many different values can C have?**

a) 2 b) 3 c) 4 d) More than 4

---

4. **56 × k is a perfect cube where k is a natural number. What could be the smallest possible value of k?**

a) 36 b) 49 c) 56 d) 72

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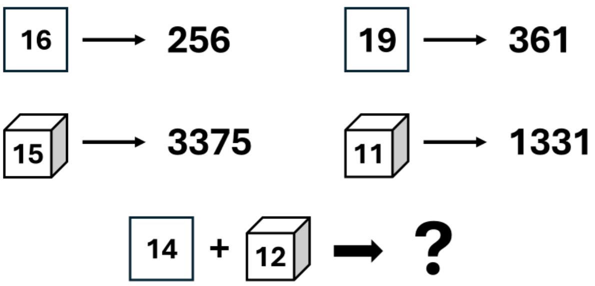
5. **Each geometrical shape denotes a certain operation. What will come in place of "?"**



a) 2888 b) 1914 c) 1924 d) 340

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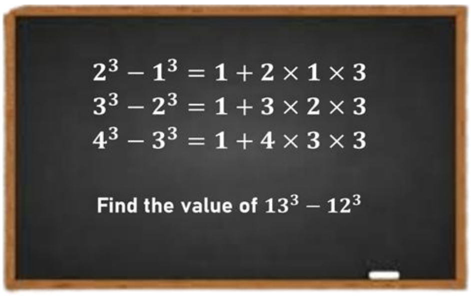
6. **XYZ is a 3-digit number such that it is the square of a multiple of 5.**
What will be the HIGHEST possible remainder of (XYZ)/100?

a) 10 b) 15 c) 20 d) 25

7. **A class teacher wrote the following pattern on the board, where an expression is written after the "=" sign in each row. She then asked the class to find the expression for 13^3 - 12^3. Which DIGIT appears the HIGHEST number of times in the expression that denotes 13^3 - 12^3?**



a) 1 b) 2 c) 3 d) 4

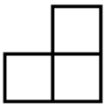
---

8. **Fill in the grid with the squares and cubes of digits 2 to 5 such that:**
   - **Each square/cube number appears twice**
   - **Both square and cube of the same number cannot appear in the same row/column**
   - **Same numbers do not appear diagonally**

   What is the maximum possible sum that can be obtained by the cells present in the configuration of Shape A (without rotation)?

64			4
27	125		
8		9	25
		16	

GRID

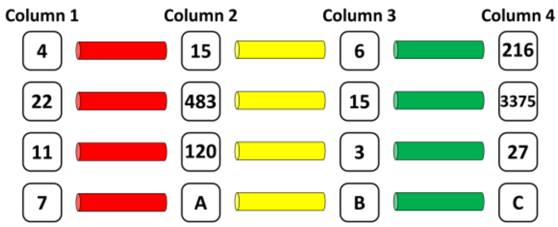


Shape A

a) 197 b) 161 c) 193 d) 216

---

9. **The image below shows a logic machine, where numbers move from Column to Column (starting from column 1 as input and reach column 4) through the tunnel, where they change in a different way, each time. What would be the sum of A, B, and C?**



a) 2259 b) 1795 c) 180 d) 1788

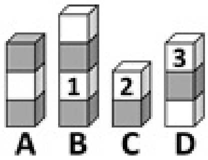
---

10. **Every column follows a certain rule. What number should come in place of "?"**

CL1	CL2	CL3	CL4
16	28	25	12
9	7	16	27
12	14	20	?

a) 17 b) 18 c) 19 d) 21

Boxes are stacked in four columns A, B, C, and D, such that:
• Each box is labelled with a number from 1, 2, and 3, with labels on some of the boxes already shown in the image given below
• No two adjacent boxes in the same column are labelled with the same number
• No two adjacent columns have the same number on the topmost box
• For every column, the sum of the numbers labelled on the topmost and bottommost block is equal
In the shaded boxes, which of these numbers will occur the HIGHEST number of times?



a) 1 b) 2 c) 3 d) All of them occur equally

Chapter 2: Power Play

1. **Each term is written in exponential form. If the terms are rearranged in ascending order of their numerical values from left to right, how many terms remain in the same position as in the original arrangement?**

   Left $4^3,\ 2^4,\ 3^3,\ 2^8,\ 5^2,\ 3^6$ **Right**

   a) 0 b) 1 c) 2 d) 3

---

2. **What will come in place of "?" in the given series?**

   6, 26, 126, 626, ?

   a) 3125 b) 3126 c) 926 d) 916

---

3. **A team is to be formed from a group of 7 students: A, B, C, D, E, F, and G where:**
   - **A and B cannot be in the same team, but at least one of them must be included in the team**
   - **C and D must either both be included or both be excluded from the team**
   - **E and F must either both be included or both be excluded from the team**

   If the team consists of 4 students, in how many different ways can it be formed?

   a) 2 b) 4 c) 3 d) 6

---

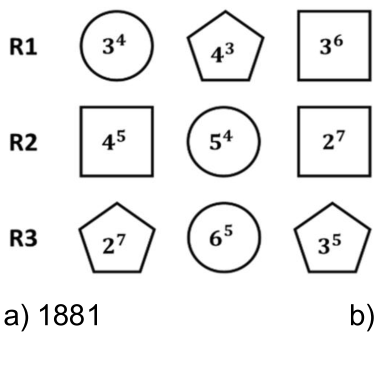
4. **Let X, Y, and Z be single-digit whole numbers. The number XY is a two-digit number formed using digits X and Y.**
   If 4000 < $(XY)^Z$ < 5000, what is the minimum possible value of XY?

   a) 10 b) 16 c) 17 d) 15

---

5. **Raj must select three numbers from the grid from three DISTINCT shapes such that:**
   1. **Each number is chosen from a different row**
   2. **The number selected from Row R1 is the nth power of 3**
   3. **No digit is repeated among the bases and exponents of the chosen numbers**

   Based on these rules, what is the sum of the three selected numbers?



   a) 1881 b) 7968 c) 1482 d) 1100

6. A man has a bag with a maximum capacity of 2^4 units. He earns certain points for the books he carries in that bag and the load of each book is shown in the table below:
   - For every Management book added to the bag, 2 Fiction books must be added
   - For every Mathematics book added to the bag, 2 Physics books must be added

   If he has to carry at least one book of each genre, what is the maximum number of points he can earn without exceeding the bag's capacity?

BOOK GENRE	LOAD	POINTS
Management	2^2	2^5
Mathematics	2^1	2^4
Physics	2^0	2^3
Fiction	2^0	2^2

a) 116　　b) 120　　c) 112　　d) 104

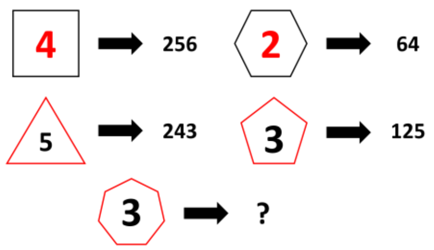
7. How many possible combinations of 1 owl and 1 tree can be made from 3 owls and 5 trees?

a) 9　　b) 10　　c) 15　　d) 16

8. Sam has Rs. 4000 in such a way that he has Re. 1 coins in power of 10, Rs. 2 coins in power of 5, Rs. 5 coins in power of 2, and the remaining coins of Rs. 10. The amount of money of each individual denomination of Re. 1, Rs. 2, and Rs. 5 is greater than or equal to thousand. At MAXIMUM, how many coins of Rs. 10 can he have?

a) 53　　b) 47　　c) 38　　d) 45

9. What will come in place of "?"



a) 243　　b) 125　　c) 343　　d) 2187

10. Sam and Tim are playing a number-maximizing game. Their numbers are shown in the table below (where A and B are undefined). They must modify their numbers using ONLY ONE of the following rules:
   - If the base is smaller than the exponent, the base n is replaced with 1/n
   - If the base is greater than the exponent, the exponent n is replaced with 1/n

   Sam and Tim choose A and B and apply the rule exactly once in a way that maximizes their own numbers. What is the difference between the final numbers of Sam and Tim, where A and B are single-digit natural numbers?

Sam's number	4^A
Tim's number	B^2

a) 2　　b) 3　　c) 1　　d) 0

Enter all the letters and the numbers of the Set in the empty squares of the grid given below, such that:
- Every vowel must have an even number in at least one of its adjacent squares
- The letter 'H' is not adjacent to 'A' or 6
- Two consecutive numbers cannot be in any adjacent squares
What will come in place of “?”
Note: *Squares are considered to be adjacent only if they share a common side. Squares sharing a common corner are not considered adjacent*

	5	
3	?	H

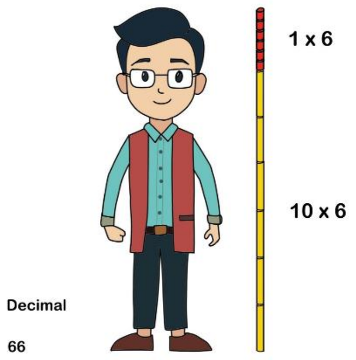
Set: A, E, Y, 2, 4, 6

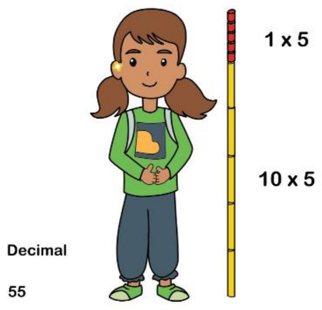
a) 6 b) Y c) E d) A

Chapter 3: A Story of Numbers

Activity Time

Measure your Height





Activity: Measuring your Height

1: Measuring your height with Decimal Pipes

We generally measure our height in the decimal number system, i.e., using base -10 numbers. We use powers of ten 10^0, 10^1, 10^2, 10^3, … (1, 10, 100, 1000, …) and digits from 0 to 9 (ten distinct symbols), to represent any number that exists.

Imagine now that you are measuring heights using PVC pipes of fixed lengths:
1 inch, 10 inches, 100 inches, and so on. Each pipe size is a power of 10. You also have 9 pipes of each length.

1. **How many of the decimal pipes will it take to measure the height of a 66-inch-tall person?**
   a) 21 pipes: 5 pipes of 10 inches height and 16 pipes of 1 inch height
   b) 7 pipes: 7 pipes of 10 inches height
   c) 9 pipes: 6 pipes of 10 inches and 3 pipes of 2 inches height
   d) 12 pipes: 6 pipes of 1 inch and 6 pipes of 10 inches

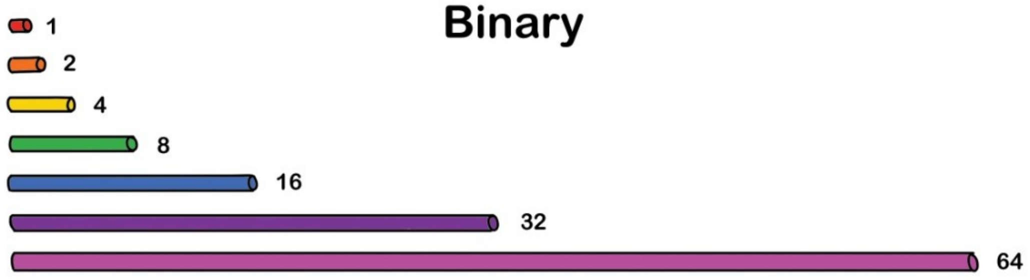
2: Measuring your height with binary pipes

We keep hearing that the language of computers is binary, the language of 0 and 1. But what are these binary numbers? How do we understand these in light of what we already know?

Instead of using decimal pipes, which are powers of 10, what happens if we use pipes of length 1, 2, 4, 8, 16, 32, 64, and so on? What is special about these numbers?

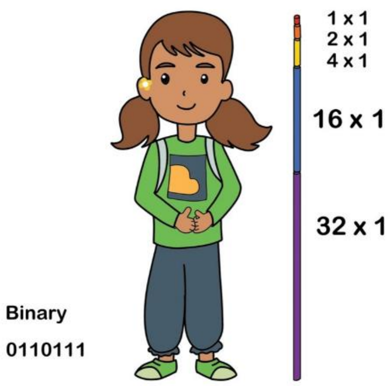
They are powers of 2. We have exactly one copy of each pipe length. We will call these pipes Binary pipes**.**

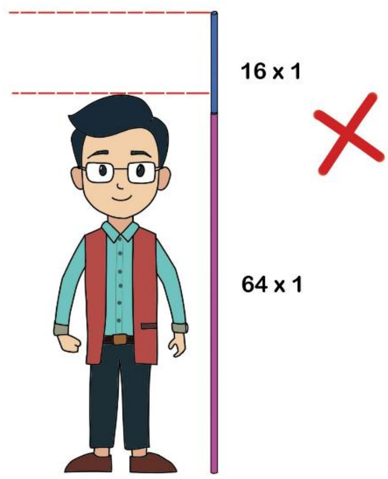
Binary



Let's measure a height in binary. For example, 55 = 32 + 16 + 4 + 2 + 1. We can tabulate the information below. Using this, we can say that the binary representation of 55 is $(110111)_2$.

55 in binary	2^5=32	2^4=16	2^3=8	2^2=4	2^1=2	2^0=1
$(110111)_2$	1	1	0	1	1	1





1. **Which all binary pipes will we use to measure the height of a 66-inch-tall person?**
   a) 63, 3
   b) 32, 16, 8, 4, 2, 1, 2, 1
   c) 64, 2
   d) 32, 32, 2

---

2. **Binary representation of 10 is:**
   a) 1010 b) 1111 c) 1000 d) 1100

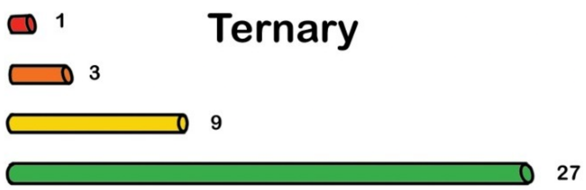
---

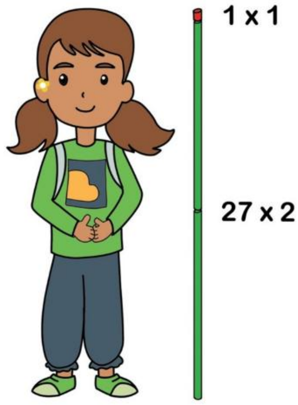
3. **Why can't we use the digit "2" in the representation of a number in binary?**
   a) It is illegal, as only 0 and 1 are given to us in the rules
   b) We can write any number using 0 and 1; there is no need for 2
   c) Computers dislike it
   d) It is too big

3: Measuring your height with ternary pipes

Instead of base 10 (decimal) and base 2 (binary), what if we set the base to something else, for example, 3?

Then we will have pipes of lengths 1, 3, 9, 27, 81…, all powers of 3, and 2 pipes of each length.





55 inches is 54 + 1 = 2 x 27 + 1 x 1.

55	3^3	3^2	3^1	3^0
$(2001)_3$	2	0	0	1

1. **When we measure 66 inches with ternary pipes, how will its representation be?**
   a) 2011 b) 2101 c) 2110 d) 2210

2. **What is the largest digit allowed in Base 3?**
   a) 3 b) 2 c) 1 d) 9

3. **How will we write 100 in base 3 (ternary) notation?**
   a) $(3201)_3$ b) $(2333)_3$ c) $(11000)_3$ d) $(10201)_3$

Conclusion

We see that a number can be represented in many forms and symbols. Decimal notation is one of the many possible base number systems (Positional value system).

The fundamental idea is that we arrange the powers of the base number n in increasing order from right to left.

The face value at each place tells us how many times the corresponding power of the base is used to represent the number.

We also see that the base n system requires n distinct digits. These digits are 0 till n - 1.

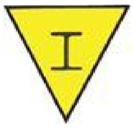
Questions

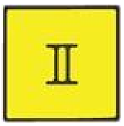
1. **Some terms are missing in between the sequence. Find the missing terms in the order from left to right. VII, VI, VI, VII, V, VIII , ______ , _______ , III , X**

   a) VI, IX b) V, VIII c) IV, IX d) IV, V

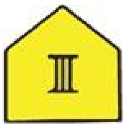
---

2. **Find the odd one out from the following:**

   a) 

   b) 

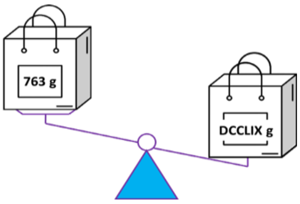
   c) 

   d) 

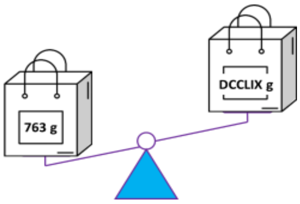
---

3. **In each of the given options, the weight of one bag is written in the Roman numeral form and that of the other bag is in Hindu - Arabic form. Identify the option that displays the tilt of the balance correctly.**

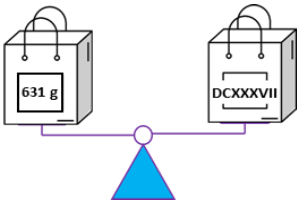
   Note: *The balance tilts towards the larger weight*

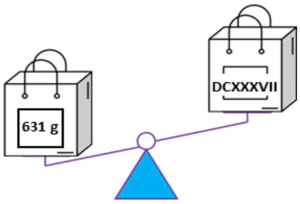
   a)

   b)

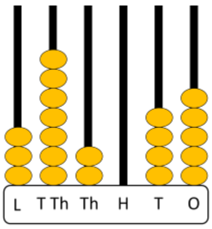
   c)

   d)

---

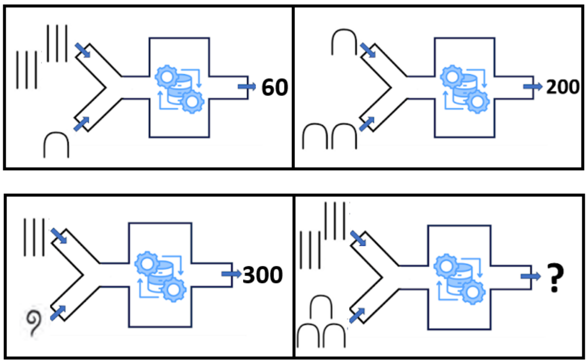
4. **On the abacus, select two poles, A and B, where pole B has fewer beads than pole A. Exactly one bead from pole A must be moved to pole B.**
   If more than one pole has fewer beads than A, move the bead to the pole that has the maximum number of beads among those poles.
   After making this single move, what is the highest possible 6-digit number that can be formed?

   a) 3,62,046 b) 4,72,035 c) 4,72,046 d) 4,62,045

5. What will come in place of "?"

Note: *All the numerals on the left of each term are from the same number system*



a) 18 b) 90 c) 36 d) 180

---

6. A 3 - digit code is formed using Chinese numerals (Zongs) and Roman numerals. Each row represents a guess, along with a statement describing how correct the guess is. Using these clues, determine the correct code.

IX	V	𝍦	One digit is correct but incorrectly placed and incorrect numeral
𝍧	𝍠	𝍥	One digit is well placed but wrong numeral
III	VIII	𝍡	No digit is correct
𝍦	VI	𝍤	Two digits in the right numerals are correct but in the wrong positions.
𝍡	𝍢	IV	One correct digit at incorrect position and incorrect numeral

a) V 𝍣 𝍥 b) 𝍤 𝍣 VI c) 𝍣 𝍦 VI d) 𝍣 𝍤 𝍥

---

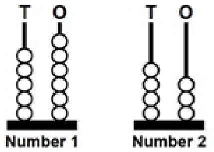
7. If certain numbers are coded as shown in the image, what would be the code for 208?

Number	Code
1	I
3	○
12	△○
34	□○○I
87	▽○○

a) ▽▽□△△I b) ▽▽□△△○

c) ▽▽□△○I d) ▽▽△△○I

8. **Two 2-digit numbers are represented below by stacking balls on poles. If you can shift/move EXACTLY ONE ball from a pole to any of the other three poles, what will be the MINIMUM possible difference between the numbers, finally?**

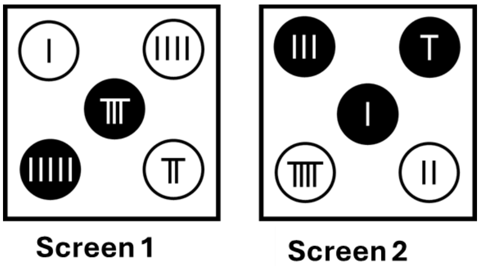


a) 1 b) 2 c) 11 d) 13

---

9. **Sam wants to create a 4-digit password using the buttons (Chinese numerals - Zongs) from the given screens. He must press the buttons in the same sequence as he wants the digits in the password to appear.**
   - **Every next button pressed must have a greater digit than the previous button**
   - **No two adjacent digits in the password can be selected from the same screen and from buttons of the same colour**

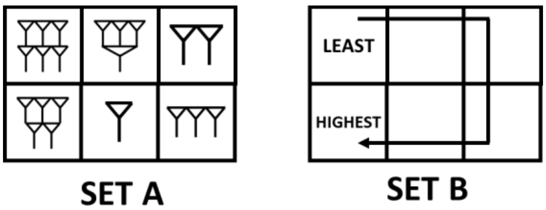
   How many different passwords can Sam form?



a) 1 b) 2 c) 3 d) 4

---

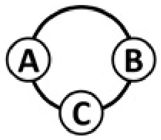
10. **Rearrange the numbers in Set A into ascending order, as shown in Set B. The blocks can be rearranged only by swapping adjacent blocks. What is the MINIMUM number of swaps required to do this?**
    Note: *Blocks that have common sides are considered to be adjacent. Blocks that have a common corner alone are not adjacent*

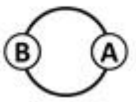


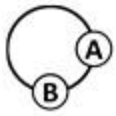
a) 3 b) 4 c) 5 d) 6

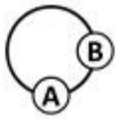
The Thinking Spot

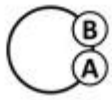
As shown below, A, B, and C are positioned at different starting points on a circular track. B runs at half the speed of A, while C runs at twice the speed of A.
If all three start running at the same time and in the same direction, what could be the positions of A and B by the time C completes one full round of the track?



(a) 

(b) 

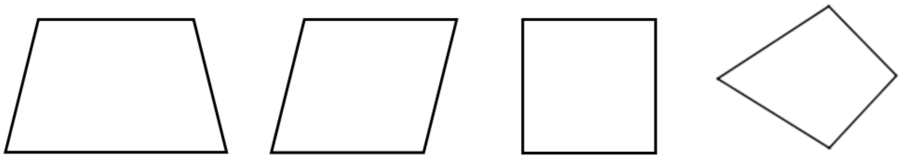
(c) 

(d) 

Chapter 4: Quadrilaterals

1. **Raj colored the following shapes (Trapezium, Parallelogram, Square, Kite) in Red, Yellow, Green, and Blue, not necessarily in the same order.**
   - **The shape colored in Red has no parallel sides**
   - **The shape with each of its angles equal to 90º, is not Yellow**
   - **The shape with two pairs of parallel sides is not Green**

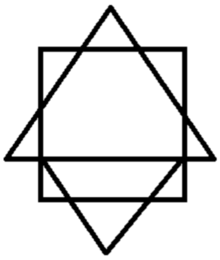
   Which shape did he color in yellow?



a) Trapezium b) Parallelogram c) Square d) Kite

---

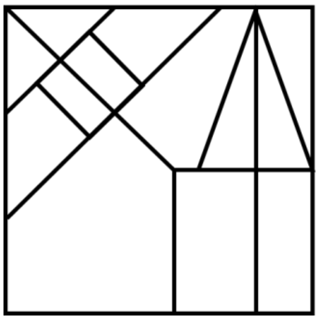
2. **How many quadrilaterals are there in the given figure?**



a) 8 b) 9 c) 10 d) 11

---

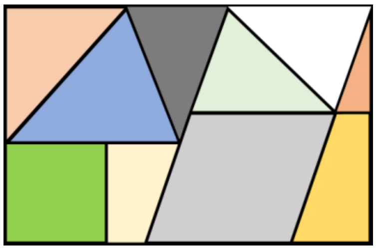
3. **How many rectangles are there in the given figure?**
   Note: *For the purpose of this question, please count all squares also as rectangles*



a) 8 b) 9 c) 10 d) 11

---

4. **In the given tangram, how many quadrilaterals are made with exactly 3 colored tiles?**



a) 4 b) 3 c) 2 d) 1

5. **Below are three families of shapes and a set of Property Cards.**
   - **A property card should be placed into a family if the property applies to that family**
   - **A single property may belong to more than one family**

   After placing all the property cards, which family will have the maximum number of properties?

Families
Rectangle Family
Square Family
Parallelogram Family

   Property Cards

   - All sides are equal and all angles are 90°.
   - All angles are 90° and opposite sides are equal.
   - Opposite sides are parallel.
   - Diagonals are equal and bisect each other.
   - Opposite sides are equal and parallel.

   a) Square Family b) Rectangle Family c) Parallelogram Family d) Both a and c

---

6. **In the grid, each letter stands for the initial of a shape: Square, Rhombus, Kite, Trapezium, and Parallelogram. How many pairs of adjacent letters represent two shapes in which the diagonals bisect each other?**

R	K	S	P	T	R	P	S	T	P	R	S	K

   a) 3 b) 4 c) 5 d) 6

---

7. **Find the option that does NOT belong to the elements in the given set.**
   Set = (Angle between diagonals of a Rhombus, Each angle of a rectangle, Each angle of a square)

   a) Angle inscribed in a semicircle
   b) Half of sum of opposite angles of a Cyclic Quadrilateral
   c) Each angle of an Equilateral Triangle
   d) Angle opposite to Hypotenuse in a Right - Angled Triangle

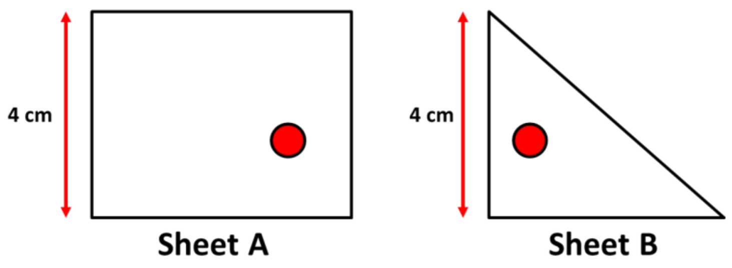
---

8. **Which shapes from the given options will NOT form a quadrilateral when any two identical shapes are joined along any one of their sides?**
   Note: *You can rotate the shapes but they cannot overlap each other*

   a) Triangle b) Rhombus c) Trapezium d) Pentagon

---

9. **Two transparent sheets are shown below. Sheet A is a rectangle and Sheet B is a right-angled triangle. Rearrange the sheets so that the red points overlap exactly (without rotating or flipping the sheets). How many quadrilaterals are formed in the final figure?**



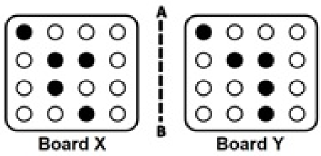
   a) 3 b) 4 c) 5 d) 6

10. **Paul and Sam are at different positions and start walking in opposite directions but along parallel paths. After covering the same distance, each of them turns toward the starting point of the other person and walks straight (without taking any other turns) until they reach each other's starting points. What can we say for certain about the shape formed by BOTH their paths together?**

    a) The shape of the paths is a rectangle
    b) It has no lines of symmetry
    c) Opposite sides are of same length
    d) Both options a and c

The Thinking Spot

Given below are two boards, X and Y. Clicking any circle on Board Y changes its colour from black to white, or white to black.
What is the MINIMUM number of clicks required to transform Board Y into the MIRROR IMAGE of Board X?



(a) 2 (b) 3 (c) 5 (d) 4

Chapter 5: Number Play

1. **A, B, C, D, and E are five distinct whole numbers, with 16 as the smallest number. When arranged in ascending order, the difference between any two consecutive numbers is 8.**
   - **It is given that D is the greatest number and A is the least**
   - **Also, B is greater than E but less than C**

   Which of the following numbers is definitely divisible by 32?

   a) A b) B c) C d) E

---

2. **A control panel has three bulbs - A, B, and C. Each bulb is associated with one fixed divisor (greater than 3). A bulb glows whenever the entered number is divisible by its own divisor. A technician tested the panel with several numbers and noted which bulbs glowed, as shown in the table below.**

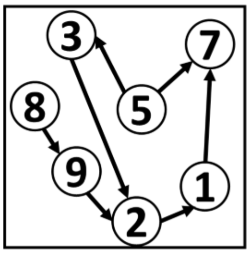
   Using this information, if 56 is entered, which bulb(s) will glow?

Number entered	Bulbs that glow
12	A,B
16	B
35	C
42	A,C
18	A

   a) A and C b) Only B c) B and C d) A and B

---

3. **Using the digits shown in the Box, form the largest possible 6-digit number that is divisible by 9 (without repeating the digits). In the number formed, which place values contain digits that point to a greater digit than themselves in the box?**



BOX

   a) Hundreds and Ones b) Ones and Tens c) Thousands and Tens d) Ones and Thousands

---

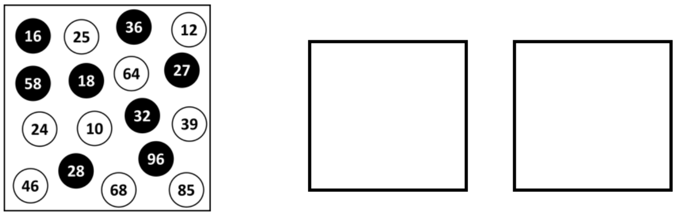
4. **Sam has two different two-digit numbers such that:**
   - Both numbers are divisible by 9
   - None of the numbers is a multiple of 18

   What is the LEAST possible sum of the two numbers?

   a) 45 b) 72 c) 63 d) 54

5. **The numbers of the SET will be distributed to the Box 1 and Box 2, based on the conditions given below:**
   - **If the number is a multiple of 4 it goes to Box 1**
   - **If the number is a multiple of 6 it goes to Box 2**
   - **If the number is a multiple of both 6 and 4 it goes in both the boxes**

   Find the ratio of the number of BLACK CIRCLES in Box 1 to the number of WHITE CIRCLES in Box 2, finally.



SET **Box 1** **Box 2**

   a) 2:1 b) 3:2 c) 5:2 d) 5:3

---

6. **Ashish and Devika together choose two different numbers from 2 to 24 (both inclusive). The chosen numbers are multiplied, and the product is divided by 25. The remainder decides the result:**
   - **If the remainder is less than 10, Devika wins**
   - **If the remainder is more than 14, Ashish wins**
   - **In all other cases, the game is a draw**

   What is the minimum possible sum of the numbers chosen by Ashish and Devika such that Ashish wins the game?

   a) 7 b) 8 c) 9 d) 6

---

7. **On the six faces of a die, even numbers from 2 to 12 are written.**
   - **All multiples of 4 are written on adjacent faces such that no two of these faces are opposite to each other**
   - **The sum of the numbers on any two opposite faces is always greater than 10**

   Which of these pairs are written on opposite faces?

   a) 8 and 10 b) 12 and 6
   c) 2 and 12 d) Cannot be determined

---

8. **In the year 2020, Simran's age was a multiple of 6. In the year 2024, her age was a multiple of 11. Which of the following could possibly be the age of Simran in the year 2024?**

   a) 11 years b) 22 years c) 33 years d) 44 years

---

9. **The middle number of 5 consecutive even numbers is 5p. Exactly two numbers in the sequence are divisible by 4, and only one is divisible by 5. Among the following values, which could be the possible value of p.**

   a) 2 b) 3 c) 4 d) 5

10. **A cryptarithmetic multiplication is shown below. Here, P, Q, M, and N are distinct single-digit natural numbers. MN represents a two-digit number, and PQN represents a three-digit number. What is the smallest possible value of P + M?**

$$\begin{array}{r} MN \\ \underline{\times \quad 6} \\ PQN \end{array}$$

a) 3 b) 1 c) 4 d) 5

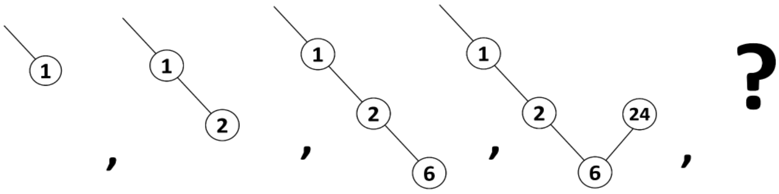
The Thinking Spot

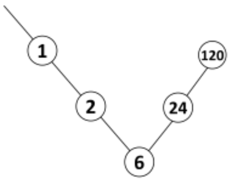
There are 9 switches in a row on a switchboard. 3 of them belong to lights and the remaining are of fans.
Every two consecutive light switches have exactly two fan switches between them. The switch at the extreme right is NOT a light switch and the switch at the extreme left is NOT a fan switch.
Which of them is definitely a light switch?

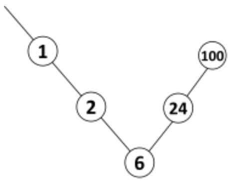
(a) 2nd switch from the left
(b) 3rd switch from the left
(c) 6th switch from the right
(d) 5th switch from the left

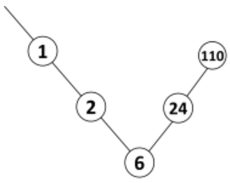
Chapter 6: We Distribute Yet Things Multiply

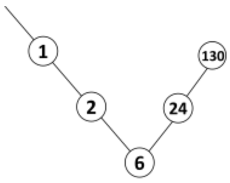
1. What will come in place of "?"



a) 

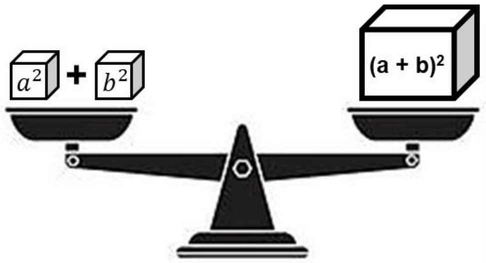
b) 

c) 

d) 

---

2. On a weighing scale, both a and b are whole numbers. Which conclusion is logically correct?



a) The scale can balance only when one of the numbers is 0
b) The scale can balance for any values of a and b
c) The scale can balance only when a = b = 1
d) The scale can balance only when a and b are single - digit natural numbers

---

3. There are 49 sets of marbles, numbered from 1 to 49. The set numbered k contains k marbles. The marbles in set 49 are sold at ₹1 per marble, those in set 48 at ₹2 per marble, those in set 47 at ₹3 per marble, and so on, with the price per marble increasing accordingly. What is the difference between the maximum and minimum total amount for which a set of marbles can be sold?

a) $49^2 - 25^2$ b) $25^2 - 7^2$ c) $49^2 - 7^2$ d) $50^2 - 25^2$

---

4. A merchant has ($19^2 - 1$) coins. He must distribute the coins to some children.
The number of children is equal to the number of distinct prime factors of 30.

- **First, each child receives 4 packs of 6 coins each**
- **Then, after this distribution, the merchant must give each child 3 more packs of 6 coins each, ONLY IF the remaining coins are more than half of the original count**

Which of the following expressions represents the number of coins finally left with the merchant, after the distribution(s)?

a) $(19^2 - 1) - 3(6 \times 4)$
b) $(19^2 - 1) - 3(6 \times 4 + 3 \times 6)$
c) $(19^2 - 1) - 3(6 \times 4 - 3(3 \times 6))$
d) $(19^2 - 1) - 5(6 \times 4 + 3 \times 6)$

5. **A two-digit number XY is formed using two different digits. When the sum of its digits is added to the number, a new number is formed, which is divisible by 3. Which of the following will be DEFINITELY true about the new number?**

a) X is a multiple of 3   b) Y is a multiple of 3
c) X + Y is a multiple of 3   d) None of these

---

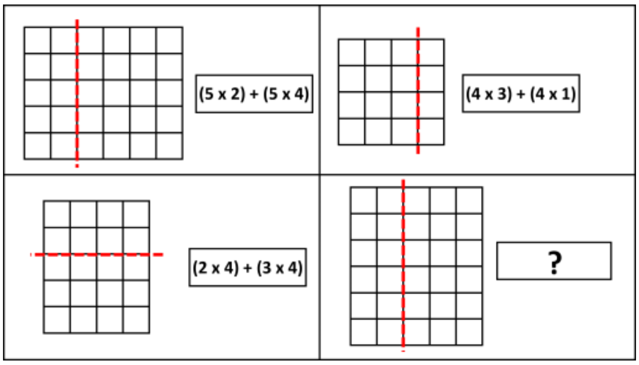
6. **A, B, and C are three different digits among 1 - 9. How many digits CANNOT be assigned to "A" if the following condition is true?**

$$A + B \times C = 25$$

a) 3   b) 4   c) 5   d) 6

---

7. **What will come in place of “?”**



a) (2 x 4) + (2 x 6)   b) (6 x 3) + (6 x 4)   c) (1 x 6) + (1 x 4)   d) (6 x 2) + (6 x 3)

---

8. **A team of three painters - A, B, and C - must paint rooms and each painter paints 3 rooms per working day. They are scheduled to work for 50 days starting Monday.**
● Painter A does not work on weekends (no Saturdays or Sundays)
● Painter B does not work on every 5th day (Day 5, Day 10, Day 15, and so on)
● Painter C works on all 50 days
Based on the above rules, determine an expression that represents the total number of rooms painted by all three painters combined.

a) 3 × 50   b) 3 × (3 × 50)   c) 3 × (36 + 40 + 50)   d) 3 × 3 × (36 + 40 + 50)

---

9. **A train runs from A to D via B and C. The number of passengers entering and leaving at each station is given in the table. Each passenger buys a separate ticket for each segment travelled (A→B, B→C, C→D).**
Ticket prices are:
A→B = Rs. 1, B→C = Rs. 2, C→D = Rs. 1
Which of the following is the total cost of all tickets purchased?

Stations	In	Out
A	150	-
B	60	80
C	70	50
D	-	150

a) Rs. [(150 x 1) + (70 x 3) + (60 x 2) + (80 x 1)]   b) Rs. [(150 x 1) + (70 x 2) + (60 x 1) + (80 x 1)]
c) Rs. [(150 x 1) + (70 x 1) + (60 x 2) + (80 x 1)]   d) Rs. [(150 x 1) + (70 x 3) + (60 x 1) + (80 x 1)]

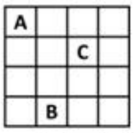
10. **If 2(pq + rs) is coded as PQ_2_RS, 3(kl + mn) is coded as KL_3_MN and 4(gh + ij) is coded as GH_4_IJ, what will be the possible code of the SIMPLIFIED FORM of the following expression? (a + b)(c + d) – (a – b)(c – d)**

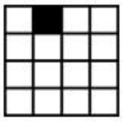
    a) AC_2_BD b) AD_3_BC c) AB_2_CD d) AD_2_BC

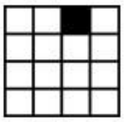
The Thinking Spot

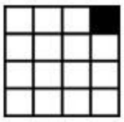
Two coins are hidden in the grid below in two different rows and columns.
- **None of the coins is in A's column, but one is in its row**
- **None of the coins is in B's row, but one is in its column**
- **None of the coins is in C's row, but one is in its column**

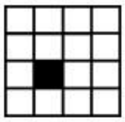
Which of the following blocks CANNOT have a coin?



a) 

b) 

c) 

d) 

Chapter 7: Proportional Reasoning

1. A fruit basket has different fruits, Bananas, Apples, Cherries, and Strawberries.
- The bananas are 1/3 the number of cherries and twice that of apples
- The strawberries are 4 more than the number of apples
- The number of fruits of each variety are all even numbers less than 15
Based on the above statement, fill in the blank with the appropriate fruit name. For every 3 strawberries, there are proportionately 6 _________.

a) Bananas b) Apples c) Cherries d) All of the above

---

2. There are two unknown proportional ratios, A : B and C : D, where A, B, C, and D are DIFFERENT single-digit numbers.
- A is 2 less than C
- D is 3 times B
What is the HIGHEST difference between any two values among A, B, C, and D?

a) 2 b) 3 c) 4 d) 5

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3. A mixture contains milk and water. When 15 litres of water is added to it, the total quantity of the mixture becomes 120 litres, where the quantity of milk and water have the LEAST POSSIBLE DIFFERENCE between them. Now, an additional 35 litres of mixture having the same milk-to-water ratio as the original mixture is added to this new mixture. What will be the ratio of milk to water in the final mixture?

a) 1:1 b) 5:6 c) 15:16 d) 16:15

---

4. Six friends A, B, C, D, E, and F are standing in a straight line positioning from 1 to 6 (from left to right).
- The positions of A and C are in 1 : 2 ratio
- E is at an extreme end and F is at an odd numbered position
- B is next to D
- The positional ratios of D to E are in proportion to C to B
What is the position of F?

a) 1 b) 3
c) 5 d) Cannot be determined

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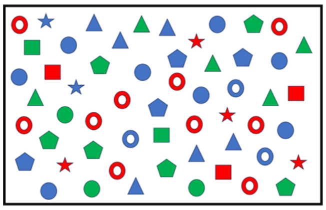
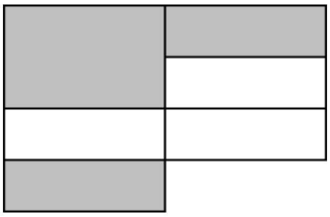
5. In the image given below, the ratio of blue stars to red stars is 1 : 2. Similarly, answer the following question based on the image.
What should replace the question mark so that the ratios are in proportion?



a) 
b) 
c) 
d) 

6. **Find the ratio of unshaded (white) rectangles to ONLY PARTIALLY shaded (grey + white) rectangles in the figure given below.**
Note: *Please count all squares as rectangles for the purpose of this question*



a) 1 : 2 b) 5 : 6 c) 5 : 7 d) 4 : 7

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7. **Three children have ages in the ratio 2 : 3 : 4. Each receives pocket money equal to Rs. 5 less than their age, and together they have Rs. 30.**
How much minimum extra money should be added so that their final amounts are in the same ratio as their ages?

a) Rs. 4 b) Rs. 6 c) Rs. 8 d) Rs. 12

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8. **In a digital clock of HH : MM format, how many times the time will be as such the HH : MM ratio will be in proportion to 1 : y where y is a single digit number?**
(Here, the clock is in 12 hour format, where HH : MM will be 00-12 : 00-59)

a) 49 b) 89 c) 81 d) 101

---

9. **A fruit seller sells apples and mangoes to 10 customers:**
   - **The price per dozen of apples and mangoes is in the ratio 3 : 5**
   - **Each customer buys a pack in which the number of apples and mangoes is in the ratio 1 : 3, and both quantities are multiples of 4**
   - **The price of each fruit is a whole number greater than 10**

   If the total amount collected from all the customers is ₹8064, what is the MAXIMUM possible number of packs sold to a customer?

a) 24 b) 18 c) 21 d) 19

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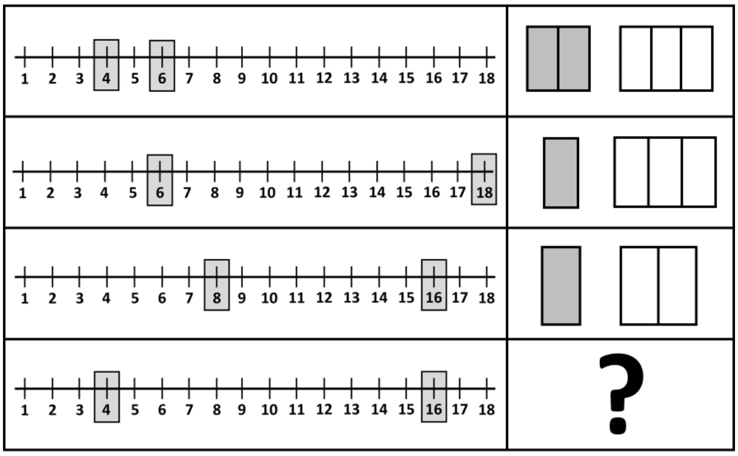
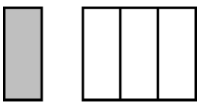
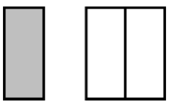
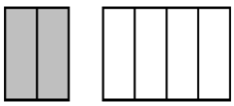
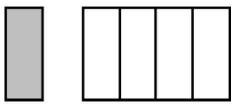
10. **What will come in place of "?"**



a)  b)  c)  d) 

The Thinking Spot

Sam arranged the following 4 blocks vertically, one on top of another, such that
• There are at least 2 blocks below the block having a circle
• The block with a triangle has at least 1 block above it
• The block having a rectangle is immediately above the block having a star
How many different arrangements satisfy all the above conditions?

(a) 0 (b) 1 (c) 2 (d) 3